Exercise 1. Exploring the lab environment

Estimated time

0:30

Overview

In this exercise, you access the environment that you use in the exercises through this course. You record variables that are provided to you by your instructor to access the environment.

Objectives

After completing this exercise, you will be able to:

- Access the VM that you use for the exercises in this course.

Introduction

In this exercise, you access the VM shared by the students in this course. You are assigned values to access services that you need for the exercises. You record the values that are assigned to you. You use the PuTTY SSH client to connect to the remote server where the software, configurations, and files for this course reside.

Requirements

- SSH client installed in your workstation. You can download PuTTY from www.putty.org.

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Exercise instructions

In this exercise, you complete the following tasks:

__ 1. Record the values for the variables to access your Hadoop cluster.

__ 2. Add the Hadoop cluster hostname to the hosts file in your local workstation.

__ 3. Access the Hadoop cluster with PuTTY.

__ 4. Install cURL

Part 1: Recording the values for the variables to access your Hadoop cluster

In this part, your instructor provides you the information that you need to access the VM and the Hadoop cluster. Record these values and use them in the subsequent exercises.

Complete the following steps:

__ 1. Copy the following table to your favorite text editor (for example Notepad) and assign the values that are provided to you by your instructor to the variables in this table.

Note

The variables listed under "**INSTRUCTOR ONLY**" are used by the instructor and *not* shared with the students.

Table 1. Variables and values for the students and the instructor in this course

Variable	Value	Description
Username	<username>	Student's username
Password	<password>	Student's password
IP address	<ip_address>	Hadoop cluster & VM IP address
Host name	<hostname>	Hadoop cluster host name
Ambari URL	<hostname:8080>	URL to access Ambari Web UI
Ambari username	<ambari username>	Username to access Ambari Web UI
Ambari password	<ambari password>	Password to access Ambari Web UI
Cluster name	<cluster_name>	Hadoop cluster name
HDFS home dir	<hdfs_home>	HDFS home directory. The value for the exercises is /user/<username>
User account home dir	$HOME	Account home directory. The value is /home/<username>. It does not need to be substituted in the exercises.
INSTRUCTOR ONLY		

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Table 1. Variables and values for the students and the instructor in this course

Variable	Value	Description
Ambari admin	<ambari admin>	Ambari administrator
Ambari admin password	<ambari admin pwd>	Ambari administrator password
Db2 Big SQL password	<bigsql_password>	**bigsql** user password
root password	<root_password>	root user password

Note

The values in the table are placeholders that are used in the exercises. When you perform each exercise, replace the place holders with the values that are provided to you by your instructor and that you recorded in the text editor.

Part 2: Adding the Hadoop cluster hostname to the hosts file in your local workstation

The Hadoop cluster hostname and domain are not added to a DNS. To resolve the hostname from the IP address you must add an entry to the hosts file in your local workstation. Update the `hostfile` with the **`<hostname>`** and **`<ip_address>`** provided by your instructor. Follow the instructions for your workstation operating system to update the hosts file.

For Windows, the hosts file is at C:\Windows\System32\drivers\etc\hosts. Follow these steps for Windows 10 workstations:

__ 1. Press the **Windows** key.

__ 2. Type **Notepad** in the search field.

__ 3. In the search results, right-click Notepad and select **Run as administrator**.

__ 4. From Notepad, open the following file. Select file type **All Files (*)**.

```
c:\Windows\System32\Drivers\etc\hosts
```

__ 5. Add the values for **`<ip_address>`** and **`<hostname>`** in the last line of the file. For example,

```
10.1.1.1 dataengineer.ibm.com
```

__ 6. Click **File** > **Save** to save your changes.

__ 7. Ping the VM by using the **`<hostname>`** that you configured in the hosts file. For example,

```
ping dataengineer.ibm.com
```

__ 8. You receive the response from the hosts as shown in the following figure.

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```
C:\>ping dataengineer.ibm.com

Pinging dataengineer.ibm.com [[redacted]] with 32 bytes of data:
Reply from [redacted]: bytes=32 time=263ms TTL=48
Reply from [redacted]: bytes=32 time=275ms TTL=48
Reply from [redacted]: bytes=32 time=256ms TTL=48
Reply from [redacted]: bytes=32 time=258ms TTL=48

Ping statistics for [redacted]:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 256ms, Maximum = 275ms, Average = 263ms
```

For Linux, the hosts file is in the directory **/etc/hosts**. Follow these steps for Linux computers:

__ 1. Open a Linux terminal. You must be logged in as **root** or as a user who can run sudo.

Add the values for **<ip_address>** and **<hostname>** in the hosts file:

```
sudo echo <ip_address> <hostname> >> /etc/hosts
```

Example:

```
sudo echo 10.1.1.1 dataengineer.ibm.com >> /etc/hosts
```

__ 2. Ping the VM by using the **<hostname>** that you configured in the hosts file. For example,

```
ping dataengineer.ibm.com
```

You receive the response from the host as shown in the following figure.

```
[root@d9035236bcbd /]# ping dataengineer.ibm.com
PING dataengineer.ibm.com (10.1.1.1) 56(84) bytes of data.
64 bytes from dataengineer.ibm.com (10.1.1.1): icmp_seq=1 ttl=37 time=20.1 ms
64 bytes from dataengineer.ibm.com (10.1.1.1): icmp_seq=2 ttl=37 time=14.6 ms
64 bytes from dataengineer.ibm.com (10.1.1.1): icmp_seq=3 ttl=37 time=13.8 ms
^C
--- dataengineer.ibm.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2011ms
rtt min/avg/max/mdev = 13.859/16.235/20.178/2.807 ms
[root@d9035236bcbd /]#
```

For Mac, the hosts file is in the directory **/private/etc/hosts**. Follow these steps for Mac computers:

__ 1. Open a Mac terminal. You must be logged in as **root** or as a user who can run sudo.

Edit the hosts file:

```
sudo nano /private/etc/hosts
```

Add the values for **<ip_address>** and **<hostname>** in the last line of the file. For example,

```
10.1.1.1 dataengineer.ibm.com
```

__ 2. Save the changes by typing **^X** (Control + X) to exist, then type **Y** if you are prompted to confirm to save the file. Press **Enter** to confirm the filename.

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__ 3. Ping the VM by using the **<hostname>** that you configured in the hosts file. For example,

```
ping dataengineer.ibm.com
```

You receive the response from the hosts as shown in the following figure.

```
MacBook-Pro:~ adelnour$ ping dataengineer.ibm.com
PING dataengineer.ibm.com (10.1.1.1): 56 data bytes
64 bytes from 10.1.1.1: icmp_seq=0 ttl=59 time=14.153 ms
64 bytes from 10.1.1.1: icmp_seq=1 ttl=59 time=17.592 ms
64 bytes from 10.1.1.1: icmp_seq=2 ttl=59 time=26.617 ms
64 bytes from 10.1.1.1: icmp_seq=3 ttl=59 time=13.174 ms
^C
--- dataengineer.ibm.com ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 13.174/17.884/26.617/5.302 ms
```

Part 3: Access the Hadoop cluster with PuTTY

In this part, you access the Hadoop cluster by using SSH client PuTTY and the variables that you recorded in Part 1. You repeat these steps in all the exercises that require PuTTY to access the Hadoop cluster.

Complete the following steps:

__ 1. Open **PuTTY** and complete the following steps:

__ 2. In the **Session** tab > **Host Name (or IP address)** enter the value for **<hostname>** provided to you by your instructor.

__ 3. In the **Connection** > **Data** tab, in **Auto-login username** and enter the **<username>** that is assigned to you and that you recorded in Part 1.

__ 4. Click **Open**.

__ 5. The PuTTY session starts. Enter the **<password>** that is provided to you by your instructor and that you recorded in Part 1 and press **Enter**.

Note

You can copy the password and paste it in the PuTTY shell by right-clicking the mouse instead of typing the password character by character.

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Information

The initial password that you enter at first login expires and you are prompted to change it for a password of your choice. When you are prompted to change your password, you must enter your initial password first and then your new password twice. Your new password must contain minimum 8 characters, which include at least one number and one uppercase letter.

After you change your password, update the “variables and values” table with the value of your new password.

Part 4: Installing cURL

If cURL is not already installed in your local workstation, download and install the version of cURL for your operating system form https://curl.haxx.se/download.html.

End of exercise

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Exercise review and wrap-up

In this exercise, you recorded the values that were provided by your instructor for several variables that you use in the exercises of this course.

You added the <hostname> and <ip_address> values to the hosts table in your local workstation.

You access the Hadoop cluster with PuTTY.

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Exercise 2. Managing Hadoop clusters with Apache Ambari

Estimated time

00:45

Overview

This exercise introduces you to the Apache Ambari Web UI. You perform basic tasks, which provide you a first-hand experience with Apache Ambari to manage your Hadoop cluster.

Objectives

After completing this exercise, you will be able to:

- Manage Hadoop clusters with Apache Ambari.
- Explore services, hosts, and alerts with the Ambari Web UI.
- Use Ambari Rest APIs.

Introduction

The Apache Ambari project is aimed at making Hadoop management simpler by developing software that enables system administrators to perform the following tasks on a Hadoop cluster:

- Provision: provides a wizard for installing Hadoop services across any number of hosts and handling configuration of Hadoop services for the cluster.
- Manage: provides central management for starting, stopping, and reconfiguring Hadoop services across the entire cluster.
- Monitor: provides a dashboard for monitoring health and status of the Hadoop cluster. Also, Ambari uses Ambari Metrics System ("AMS") for metrics collection and Ambari Alert Framework for system alerting. These functions notify you when your attention is needed.

Ambari enables application developers and system integrators to easily integrate Hadoop provisioning, management, and monitoring capabilities to their own applications with the Ambari REST APIs.

In this exercise, you start the Ambari Web UI. You explore services, hosts, and alerts by using the Ambari Web UI. Finally, you explore some of the Ambari REST APIs

Requirements

- Complete “Exercise 1. Exploring the lab environment”.

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Exercise instructions

In this exercise, you complete the following tasks:

__ 1. Start the Apache Ambari Web UI.

__ 2. Explore Services, Hosts, and Alerts with the Ambari Web UI.

__ 3. Explore the Apache Ambari REST APIs.

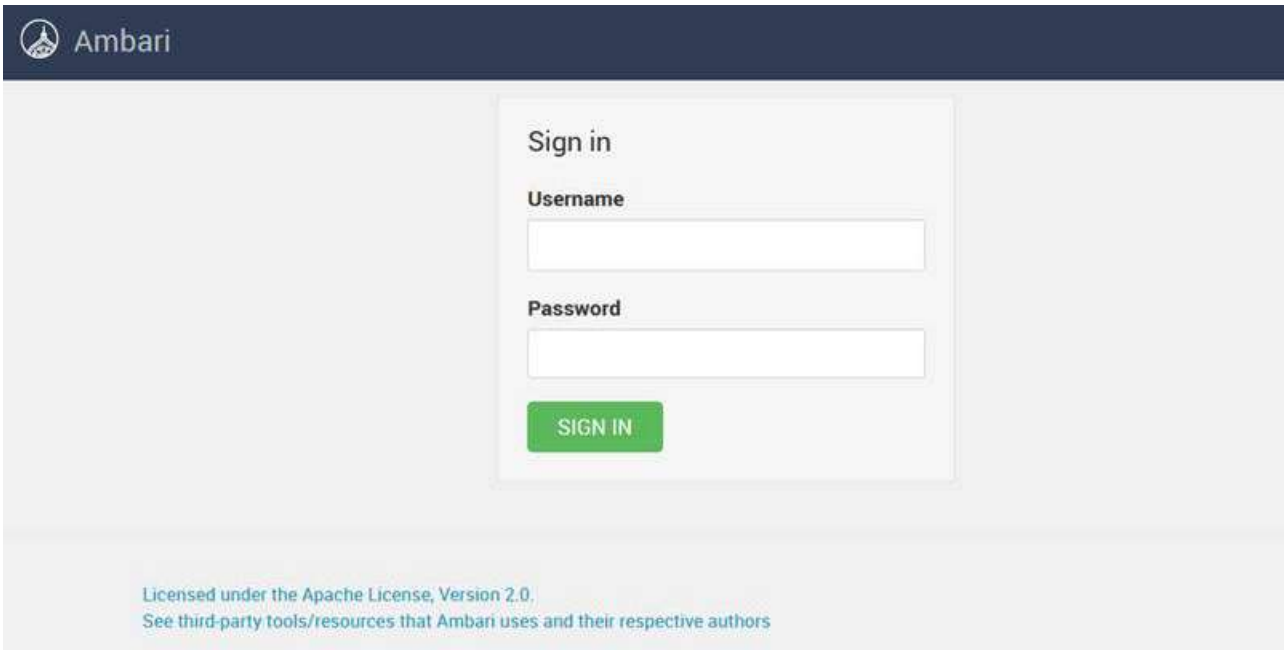
Part 1: Starting the Apache Ambari Web UI

In this part, you start the Apache Ambari Web UI, which you use to manage your Apache Hadoop cluster.

Complete the following steps:

__ 1. Enter the Ambari URL **`<hostname:8080>`** that you recorded in Exercise 1 in your browser address bar.

__ 2. Sign in to the Ambari Web UI with your Username **`<ambari username>`** and Password **`<ambari password>`** from the table in Exercise 1.



The Dashboard page is displayed.

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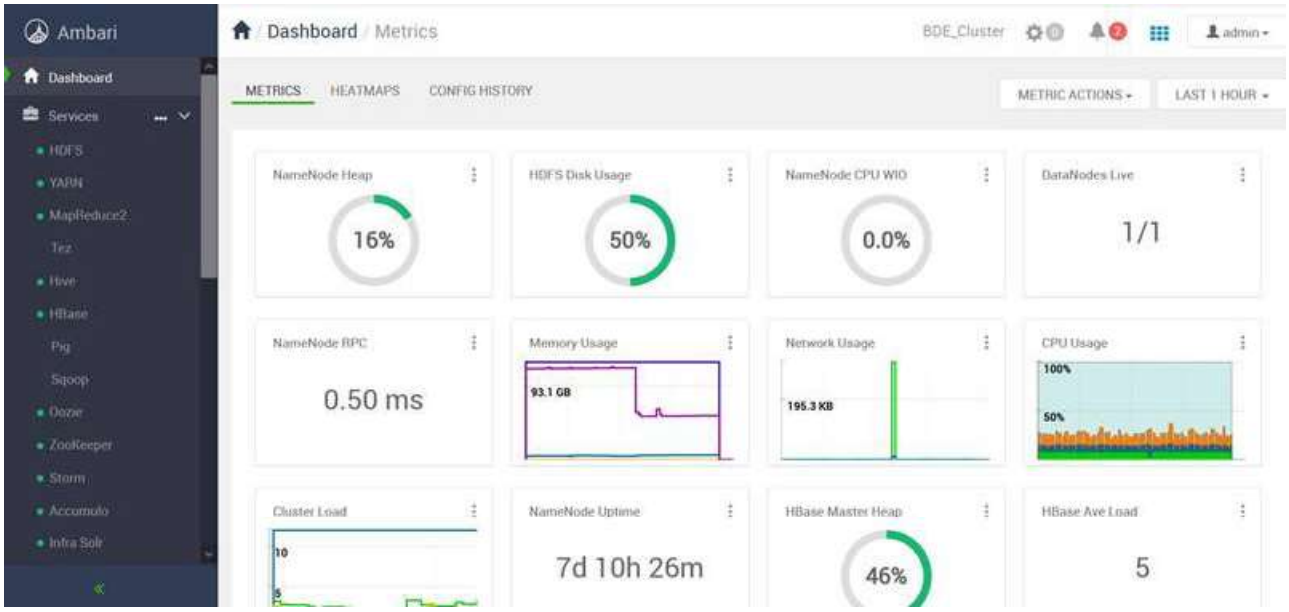

Ambari Web UI displays the Dashboard page as the home page. You can use the Dashboard to view the operating status of your cluster.

The left pane of the Ambari Web UI displays the list of Hadoop services currently running in your cluster. The dashboard includes Metrics, Heatmaps, and Config History tabs; by default, the Metrics tab is displayed. On the Metrics page, multiple widgets, represent operating status information of services in your HDP cluster.

Part 2: Explore Services, Hosts, and Alerts with the Ambari Web UI

In this part, you explore the Services, Hosts, and Alerts options in the left pane or the Ambari Web UI.

Complete the following steps:

__ 1. Click the **Hive** service that is listed under **Services** on the left pane. You see the SUMMARY and CONFIGS tabs.

__ 2. Select the **SUMMARY** tab to review overview information about the Hive service.

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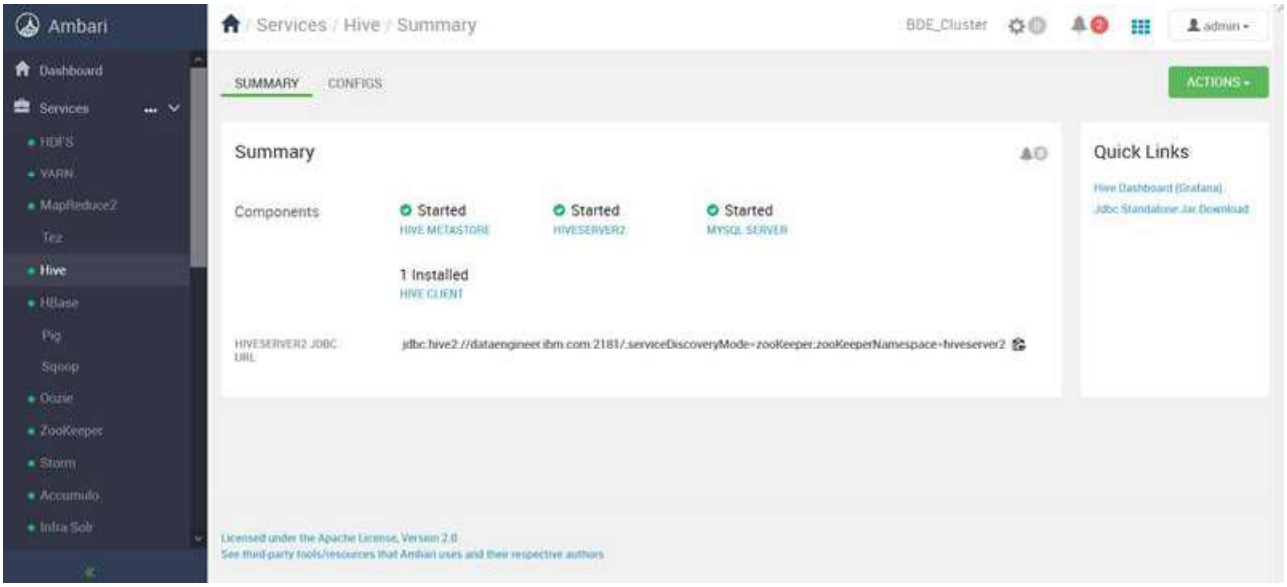
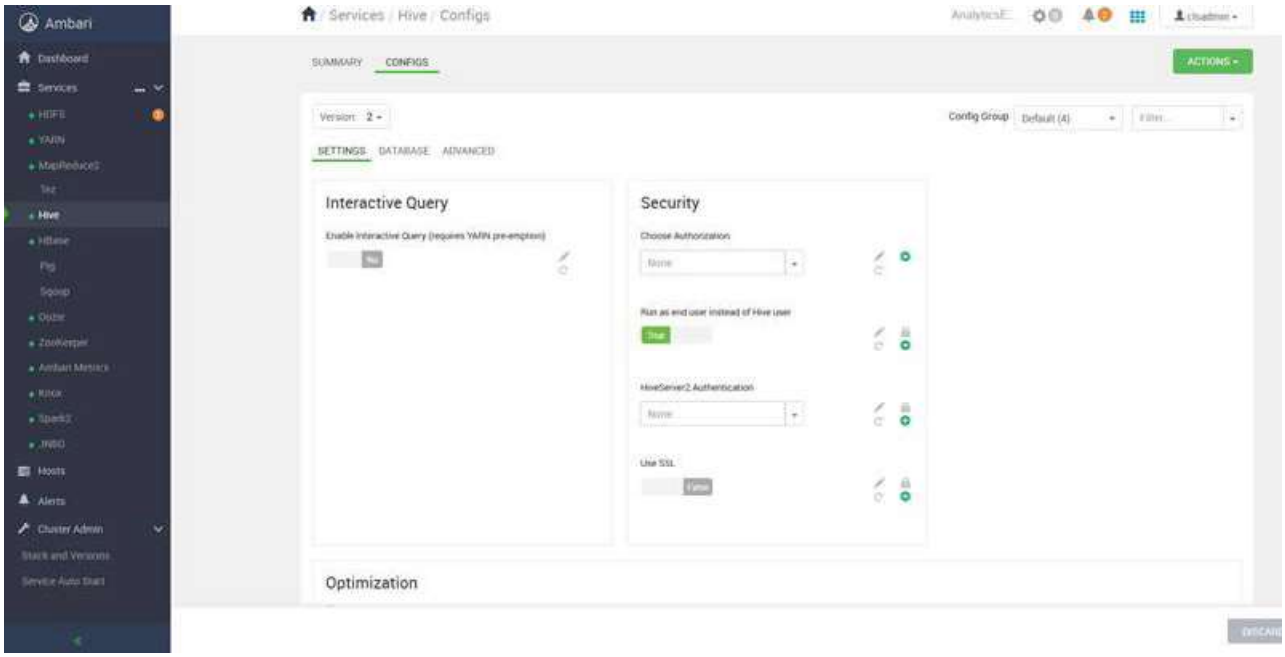


Summary displays basic information about the operational status of the selected service.

__ 3. Select the **CONFIGS** tab.



Performance of Hadoop components in a cluster can be optimized by adjusting configuration settings and property values. Changing settings and configuration values requires to log in with an Ambari admin account.

You can use the following tabs:

- SETTINGS: Manage Smart Configs to configure settings by adjusting the slider buttons.
- DATABASE: Manage database configurations
- ADVANCED: Edit advanced configuration properties and values.

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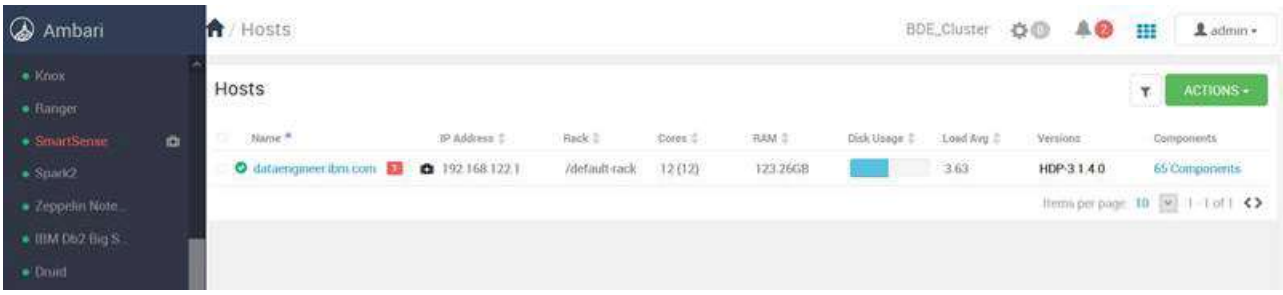
__ 4. Click **HOSTS** on the left pane.

The Hosts page is displayed



If you have an Ambari administrator account, you can use the Ambari Web UI Hosts page to manage various Hortonworks Data Platform (HDP) components, for example, DataNodes. DataNodes store data in a Hadoop cluster and it is the name of the daemon that manages the data. DataNodes run on hosts throughout your cluster. For example, you can restart all DataNode components, and optionally control this task with rolling restarts. From the Ambari Hosts page, you filter your selection of host components to manage, based on operating status, host health, and defined host groupings.

On the Hosts page, you can perform the following actions:

- Work with hosts
- Determine host status
- Filter the hosts list
- Perform host-level actions
- View components on a host
- Decommission Primary and Subordinates.

__ 5. Navigate to **Alerts** on the left pane. From the Alerts page, you can browse the list of alerts that are defined for your cluster

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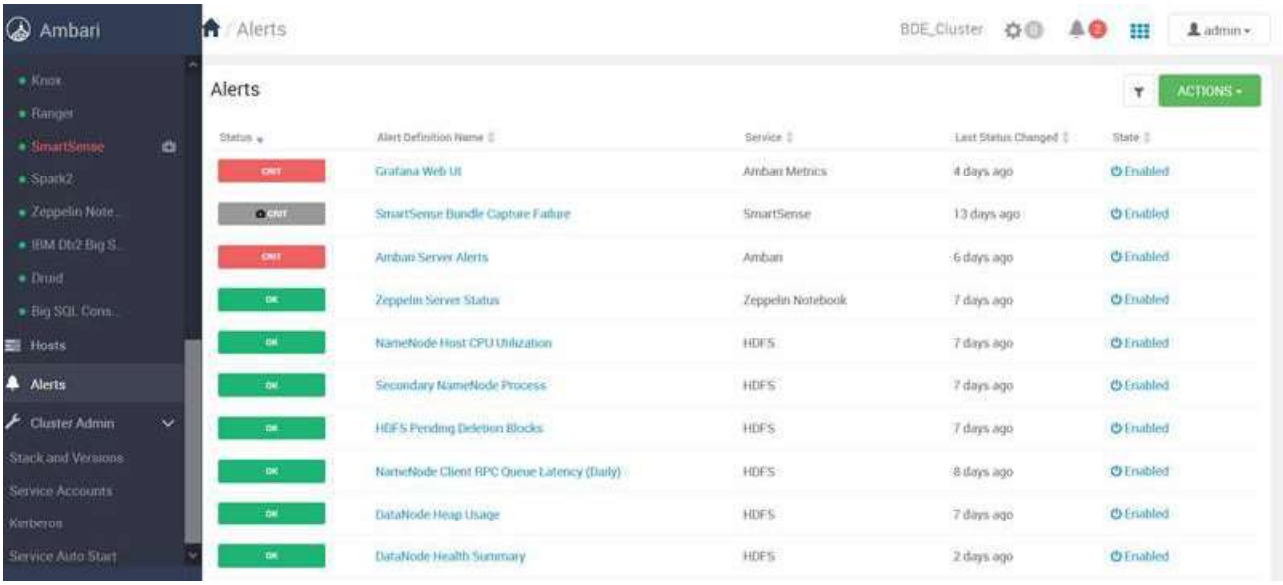

As you can see, each alert reports an alert status that indicates the alert severity. The most common severity levels are OK, WARNING, and CRIT, but there are also other severities such as UNKNOWN and NONE.

The alert statuses on your screen might be different than the ones shown in the figure.

Part 3: Exploring the Apache Ambari Rest APIs

In this part, you monitor and manage the resources in your cluster by using the Ambari REST API.

Complete the following steps:

__ 1. Open a Terminal on your local workstation.

__ 2. Run the following command in one line to view the hosts available on your cluster.

```
curl -i -u <ambari username>:<ambari password> <Ambari URL>/api/v1/hosts
```

Where **<ambari username>**, **<ambari password>**, and Ambari URL **<hostname:8080>** are the ones that you recorded in Exercise 1.

For example:

```
curl -i -u myambari_user:my_ambari_password http://dataengineer.ibm.com:8080/api/v1/hosts
```

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The results are similar to the following output:

```
HTTP/1.1 200 OK
Date: Tue, 15 Sep 2020 02:08:59 GMT
X-Frame-Options: DENY
X-XSS-Protection: 1; mode=block
X-Content-Type-Options: nosniff
Cache-Control: no-store
Pragma: no-cache
Set-Cookie:
AMBARISESSIONID=node01bp5o5ar92m111rfq6qq7ubych19.node0;Path=/;HttpOnly
Expires: Thu, 01 Jan 1970 00:00:00 GMT
User: admin
Content-Type: text/plain;charset=utf-8
X-Content-Type-Options: nosniff
Vary: Accept-Encoding, User-Agent
Transfer-Encoding: chunked

{
  "href" : "http://41.58.181.236:8080/api/v1/hosts",
  "items" : [
    {
      "href" :
"http://41.58.181.236:8080/api/v1/hosts/dataengineer.ibm.com",
      "Hosts" : {
        "cluster_name" : "BDE_Cluster",
        "host_name" : "dataengineer.ibm.com"
      }
    }
  ]
}
```

__ 3. Run the following command in one line to get details of the HDFS service in your cluster.

`curl -i -u` **<ambari username>:<ambari password> <Ambari URL>**`/api/v1/clusters/`**<cluster name>**`/services/HDFS`

Where `<ambari username>:<ambari password> <Ambari URL>` are the ones that you recorded in Exercise.

For example,

```
curl -i -u myambari_user:my_ambari_password
http://dataengineer.ibm.com:8080/api/v1/clusters/BDE_Cluster/services/HDFS
```

The results are similar to the following partial output.

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```
HTTP/1.1 200 OK
Date: Tue, 15 Sep 2020 02:12:12 GMT
X-Frame-Options: DENY
X-XSS-Protection: 1; mode=block
X-Content-Type-Options: nosniff
Cache-Control: no-store
Pragma: no-cache
Set-Cookie:
AMBARISESSIONID=node0znf5y3vpui1b1rcuqgbancva23.node0;Path=/
Expires: Thu, 01 Jan 1970 00:00:00 GMT
User: admin
Content-Type: text/plain;charset=utf-8
X-Content-Type-Options: nosniff
Vary: Accept-Encoding, User-Agent
Transfer-Encoding: chunked
{
"href" :
"http://41.58.181.236:8080/api/v1/clusters/BDE_Cluster/servi
"ServiceInfo" : {
"cluster_name" : "BDE_Cluster",
"credential_store_enabled" : "false",
"credential_store_supported" : "false",
"desired_repository_version_id" : 1,
"desired_stack" : {
"stackName" : "HDP",
"stackVersion" : "3.1",
"stackId" : "HDP-3.1"
},
"kerberos_enabled" : false,
"maintenance_state" : "OFF",
"repository_state" : "CURRENT",
"service_name" : "HDFS",
"sso_integration_desired" : false,
"sso_integration_enabled" : false,
"sso_integration_requires_kerberos" : true,
```

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```
"sso_integration_supported" : true,
"state" : "STARTED"
},
"alerts_summary" : {
"CRITICAL" : 0,
"MAINTENANCE" : 0,
"OK" : 26,
"UNKNOWN" : 2,
"WARNING" : 0
},
"alerts" : [
],
"components" : [
],
"artifacts" : [ ]
}
```


Review the list of alerts, components, and artifacts.

End of exercise

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Exercise review and wrap-up

In this exercise, you used the Ambari Web UI to perform basic tasks to gain a first-hand experience in accessing Hadoop clusters. You reviewed the Ambari Dashboard to understand the metrics that are reported on the Dashboard page. You reviewed general information about services status, configuration, and alerts. Finally, you used Ambari REST API to monitor resources.

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Exercise 3. File access and basic commands with HDFS

Estimated time

00:30

Overview

In this exercise, you explore some of the Hadoop Hortonworks directory structure on the Linux file system.

You gain experience in using the Hadoop Distributed File System (HDFS). The basic HDFS file system commands that you learn here are used throughout the remainder of the course.

In this exercise, you also move some data into HDFS. This data is used in subsequent exercises of this course.

Objectives

After completing this exercise, you will be able to:

- Describe the Hadoop Hortonworks directory structure on the Linux file system.
- Run basic HDFS commands.
- Run HDFS administrative commands.

Introduction

The general reference for HDFS commands can be found in *HDFS Commands Guide* at http://hadoop.apache.org/docs/current/hadoop-project-dist/hadoop-hdfs/HDFSCommands.html.

The HDFS commands are generally prefixed by `hdfs dfs`.

There are other HDFS commands, generally administrative commands that include parameters other than `dfs`, for example, `fsck`, `balancer`.

In this exercise, you also explore some of the Hadoop Hortonworks directory structure on the Linux file system that you use in the exercises.

Requirements

- Complete “Exercise 1. Exploring the lab environment”.
- PuTTY installed in your local workstation.

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Exercise instructions

In this exercise, you complete the following tasks:

__ 1. Explore the Hadoop directory structure on the Linux file system.

__ 2. Explore basic HDFS commands.

__ 3. Explore one of the HDFS administrative commands.

Part 1: Exploring the Hadoop directory structure on the Linux file system

In this part, you review the location of some configuration files in the Linux file system to become familiar with the Hadoop Hortonworks file system structure.

__ 1. Open **PuTTY** to connect to the VM. Enter the **<hostname>**, **<username>** and **<password>** that you recorded in "Exercise 1. Exploring the lab environment".

__ 2. Change the directory to `/usr/hdp` by running the command:

```
cd /usr/hdp
```

__ 3. List the files and directories under the directory `/usr/hdp` by running the following command:

```
ls -l
```

The results are similar to the following output.

```
drwxr-xr-x 39 root root 4096 Aug 24 21:03 3.1.4.0-315
drwxr-xr-x  2 root root 4096 Sep 28 18:29 current
drwxr-xr-x  3 root root   17 Aug 24 12:27 share
```

The first of these subdirectories (3.1.4.0-315) is the release level of the HDP software that you are working with; the version in your lab environment might differ if the software was updated.

__ 4. List the contents of the directory **`3.1.4.0-315`** (or equivalent in your environment) by running the command:

```
ls -l 3*
```

Partial results are similar to the following output.

```
[student0001@dataengineer hdp]$ ls -l 3*
total 52
drwxr-xr-x  6 root     root       76 Aug 24 12:22 accumulo
drwxr-xr-x 11 atlas    hadoop    187 Aug 24 14:32 atlas
drwxr-xr-x  2 root     root      205 Aug 24 13:50 datafu
drwxr-xr-x  8 root     root      177 Aug 24 12:48 druid
drwxr-xr-x 26 root     root     4096 Aug 24 21:02 etc
drwxr-xr-x  9 root     root     4096 Aug 24 12:37 hadoop
drwxr-xr-x  7 root     root     4096 Aug 24 12:22 hadoop-hdfs
drwxr-xr-x  6 root     root     8192 Aug 24 12:32 hadoop-mapreduce
drwxr-xr-x  9 root     root     4096 Aug 24 12:31 hadoop-yarn
drwxr-xr-x 11 root     root      133 Aug 23  2019 hbase
```

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Notice the user IDs associated with the various directories. Some have a username that is the same as the software held in the directory; some are owned by root.

__ 5. List the content of the **current** directory, which has a set of links that point back to files and subdirectories in the directory **3.1.4.0-315** by running the command:

```
ls -l current
```

```
lrwxrwxrwx 1 root root 32 Aug 24 12:22 hadoop-hdfs-nfs3 -> /usr/hdp/3.1.4.0-315/hadoop-hdfs
lrwxrwxrwx 1 root root 32 Aug 24 12:22 hadoop-hdfs-portmap -> /usr/hdp/3.1.4.0-315/hadoop-hdfs
lrwxrwxrwx 1 root root 32 Aug 24 13:50 hadoop-hdfs-secondarynamenode -> /usr/hdp/3.1.4.0-315/hadoop-hdfs
lrwxrwxrwx 1 root root 32 Aug 24 12:22 hadoop-hdfs-zkfc -> /usr/hdp/3.1.4.0-315/hadoop-hdfs
lrwxrwxrwx 1 root root 34 Aug 24 12:22 hadoop-httpfs -> /usr/hdp/3.1.4.0-315/hadoop-httpfs
lrwxrwxrwx 1 root root 37 Sep 28 18:28 hadoop-mapreduce-client -> /usr/hdp/3.1.4.0-315/hadoop-mapreduce
lrwxrwxrwx 1 root root 37 Aug 24 13:38 hadoop-mapreduce-historyserver -> /usr/hdp/3.1.4.0-315/hadoop-mapreduce
lrwxrwxrwx 1 root root 32 Sep 28 18:29 hadoop-yarn-client -> /usr/hdp/3.1.4.0-315/hadoop-yarn
lrwxrwxrwx 1 root root 32 Aug 24 13:39 hadoop-yarn-nodemanager -> /usr/hdp/3.1.4.0-315/hadoop-yarn
lrwxrwxrwx 1 root root 32 Aug 24 13:53 hadoop-yarn-registrydns -> /usr/hdp/3.1.4.0-315/hadoop-yarn
lrwxrwxrwx 1 root root 32 Aug 24 13:50 hadoop-yarn-resourcemanager -> /usr/hdp/3.1.4.0-315/hadoop-yarn
lrwxrwxrwx 1 root root 32 Aug 24 13:53 hadoop-yarn-timelinereader -> /usr/hdp/3.1.4.0-315/hadoop-yarn
lrwxrwxrwx 1 root root 32 Aug 24 12:32 hadoop-yarn-timelineserver -> /usr/hdp/3.1.4.0-315/hadoop-yarn
lrwxrwxrwx 1 root root 26 Sep 28 18:27 hbase-client -> /usr/hdp/3.1.4.0-315/hbase
lrwxrwxrwx 1 root root 26 Aug 24 13:40 hbase-master -> /usr/hdp/3.1.4.0-315/hbase
lrwxrwxrwx 1 root root 26 Aug 24 13:40 hbase-regionserver -> /usr/hdp/3.1.4.0-315/hbase
lrwxrwxrwx 1 root root 25 Sep 28 18:28 hive-client -> /usr/hdp/3.1.4.0-315/hive
lrwxrwxrwx 1 root root 25 Aug 24 13:41 hive-metastore -> /usr/hdp/3.1.4.0-315/hive
lrwxrwxrwx 1 root root 25 Aug 24 13:41 hive-server2 -> /usr/hdp/3.1.4.0-315/hive
lrwxrwxrwx 1 root root 25 Aug 24 13:06 hive-server2-hive -> /usr/hdp/3.1.4.0-315/hive
lrwxrwxrwx 1 root root 45 Aug 24 13:51 hive_warehouse_connector -> /usr/hdp/3.1.4.0-315/hive_warehouse_connector
lrwxrwxrwx 1 root root 34 Aug 24 13:06 hive-webhcat -> /usr/hdp/3.1.4.0-315/hive-hcatalog
lrwxrwxrwx 1 root root 26 Aug 24 13:41 kafka-broker -> /usr/hdp/3.1.4.0-315/kafka
lrwxrwxrwx 1 root root 25 Aug 24 13:42 knox-server -> /usr/hdp/3.1.4.0-315/knox
lrwxrwxrwx 1 root root 26 Sep 28 18:28 livy2-client -> /usr/hdp/3.1.4.0-315/livy2
lrwxrwxrwx 1 root root 26 Aug 24 13:52 livy2-server -> /usr/hdp/3.1.4.0-315/livy2
lrwxrwxrwx 1 root root 25 Aug 24 12:22 livy-client -> /usr/hdp/3.1.4.0-315/livy
```

Note

In some cases, there are multiple links to one directory. The software that is executed at this time (in the current directory) is determined by these links. Since you can have multiple versions of the same software installed, it is possible to install the software for an upcoming upgrade and later have Ambari install a rolling upgrade that systematically changes the links as appropriate to files in the new, upgraded software directories.

__ 6. Change the directory to the directory **/etc** by running the command:

```
cd /etc
```

__ 7. List the contents of the directory **/etc** by running the command:

```
ls -l
```

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Partial results are similar to the following output.

```
[student0001@dataengineer etc]$ ls -l
total 1504
drwxr-xr-x.  3 root     root        101 Aug 21 15:03 abrt
drwxr-xr-x   4 root     root         56 Aug 24 12:22 accumulo
-rw-r--r--.  1 root     root         16 Aug 21 15:10 adjtime
-rw-r--r--.  1 root     root       1518 Jun  7  2013 aliases
-rw-r--r--.  1 root     root      12288 Aug 21 15:36 aliases.db
drwxr-xr-x.  3 root     root         65 Aug 21 15:05 alsa
drwxr-xr-x.  2 root     root       4096 Aug 25 13:50 alternatives
drwxr-xr-x   3 root     root         18 Aug 24 11:31 ambari-agent
drwxr-xr-x   3 root     root         18 Aug 24 14:18 ambari-infra-solr
drwxr-xr-x   3 root     root         18 Aug 24 13:45
ambari-metrics-collector
drwxr-xr-x   3 root     root         18 Aug 24 13:47 ambari-metrics-grafana
drwxr-xr-x   3 root     root         18 Aug 24 13:46 ambari-metrics-monitor
drwxr-xr-x   3 root     root         18 Aug 24 10:41 ambari-server
drwxr-xr-x   3 root     root         18 Aug 24 13:45 ams-hbase
-rw-------.  1 root     root        541 Aug  9  2019 anacrontab
-rw-r--r--.  1 root     root         55 Aug  8  2019 asound.conf
-rw-r--r--.  1 root     root          1 Oct 30  2018 at.deny
drwxr-xr-x   4 root     root         56 Aug 24 12:36 atlas
drwxr-x---.  3 root     root         43 Aug 21 15:03 audisp
drwxr-x---.  3 root     root         83 Aug 21 15:36 audit
-rw-r--r--.  1 root     root      14781 Aug  8  2019 autofs.conf
-rw-------.  1 root     root        232 Aug  8  2019 autofs_ldap_auth.conf
-rw-r--r--.  1 root     root        795 Aug  8  2019 auto.master
drwxr-xr-x.  2 root     root          6 Aug  8  2019 auto.master.d
-rw-r--r--.  1 root     root        524 Aug  8  2019 auto.misc
-rwxr-xr-x.  1 root     root       1260 Aug  8  2019 auto.net
-rwxr-xr-x.  1 root     root        687 Aug  8  2019 auto.smb
drwxr-xr-x.  4 root     root         71 Aug 21 15:05 avahi
```

Note

Notice the long list that is displayed. You can view the configuration files for the various Hadoop components and the software that comes with the HDP installation in the directories `/etc/<componentname>/conf/*`. Do NOT try to manually change configuration files here. The configs are managed with Ambari, and you should only edit configuration settings from the Ambari Web UI.

__ 8. Take a quick look at the **/conf** directory for one of the components. Enter the following command to change the directory to Zookeeper's **conf** directory:

```
cd /etc/zookeeper/conf
```

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__ 9. List the contents of this directory by running the following command:

```
ls -l
```

The results are similar to the following output.

```
[student0001@dataengineer ~]$ cd /etc/zookeeper/conf
[student0001@dataengineer conf]$ ls -l
total 24
-rw-r--r-- 1 zookeeper hadoop  548 Aug 24 13:57 configuration.xsl
-rw-r--r-- 1 zookeeper hadoop 2476 Aug 24 13:57 log4j.properties
-rw-r--r-- 1 zookeeper hadoop  975 Aug 24 13:57 zoo.cfg
-rw-r--r-- 1 root      root   1203 Aug 23  2019 zookeeper-env.cmd
-rwxr-xr-x 1 zookeeper hadoop  311 Sep 28 18:29 zookeeper-env.sh
-rw-r--r-- 1 zookeeper hadoop  922 Aug 23  2019 zoo_sample.cfg
```

Part 2: Exploring basic HDFS commands

In this part, you explore basic HDFS commands and you move some data into HDFS.

Complete the following steps:

__ 1. In your PuTTY session, run the following command to make a directory under your **<username>** directory.

```
hdfs dfs -mkdir Gutenberg
```

__ 2. Run the following commands to upload sample files to HDFS.

```
cd /labfiles
hdfs dfs -put *.txt Gutenberg
```

__ 3. List the contents of the home directory and its subdirectories in HDFS. Run the following command.

```
hdfs dfs -ls -R /user/<username>
```

The expected output is similar to the following results:

```
-rw-r--r--   3 student0000 hdfs     421504 2020-08-18 11:37
Gutenburg/Frankenstein.txt
-rw-r--r--   3 student0000 hdfs     697802 2020-08-18 11:37
Gutenburg/Pride_and_Prejudice.txt
-rw-r--r--   3 student0000 hdfs     757223 2020-08-18 11:37
Gutenburg/Tale_of_Two_Cities.txt
-rw-r--r--   3 student0000 hdfs     281398 2020-08-18 11:37
Gutenburg/The_Prince.txt
```

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Note

In the listing of files, the following is the last file:

```
-rw-r--r--   3 student0000 hdfs     281398 2020-08-18 11:37
Gutenburg/The_Prince.txt
```

Here you see the read/write (`rw`) permissions that you usually find in a typical Linux file. The "`3`"in this example is the typical replication factor for the blocks (or "splits") of the individual files. These files are too small (last file is almost 281KB) to have more than one block (the max block size is 128MB), but each block of each file is replicated three times.

You might see "`1`"instead of "`3`"in a single-node cluster (pseudo-distributed mode). This is normal for a single-node cluster as it does not really make sense to replicate multiple copies on a single node.

Part 3: Exploring one of the HDFS administrative commands

In this part, you explore HDFS administrative commands.

Complete the following steps:

__ 1. Run the following command:

```
hdfs fsck /user/<username>/Gutenberg
```

The output is similar to the following results:

```
[student0002@dataengineer labfiles]$ hdfs fsck /user/student0002/Gutenberg
Connecting to namenode via
http://dataengineer.ibm.com:50070/fsck?ugi=student0002&path=%2Fuser%2Fstudent0002%
2FGutenberg
FSCK started by student0002 (auth:SIMPLE) from /192.168.122.1 for path
/user/student0002/Gutenberg at Fri Oct 16 16:07:43 BST 2020

/user/student0002/Gutenberg/Frankenstein.txt:  Under replicated
BP-743085664-192.168.122.1-1598275132815:blk_1073767383_26566. Target Replicas is
3 but found 1 live replica(s), 0 decommissioned replica(s), 0 decommissioning
replica(s).

/user/student0002/Gutenberg/Pride_and_Prejudice.txt:  Under replicated
BP-743085664-192.168.122.1-1598275132815:blk_1073767384_26567. Target Replicas is
3 but found 1 live replica(s), 0 decommissioned replica(s), 0 decommissioning
replica(s).

/user/student0002/Gutenberg/Tale_of_Two_Cities.txt:  Under replicated
BP-743085664-192.168.122.1-1598275132815:blk_1073767385_26568. Target Replicas is
```

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```
3 but found 1 live replica(s), 0 decommissioned replica(s), 0 decommissioning
replica(s).

/user/student0002/Gutenberg/The_Prince.txt:  Under replicated
BP-743085664-192.168.122.1-1598275132815:blk_1073767386_26569. Target Replicas is
3 but found 1 live replica(s), 0 decommissioned replica(s), 0 decommissioning
replica(s).

Status: HEALTHY
 Number of data-nodes:  1
 Number of racks:               1
 Total dirs:                    1
 Total symlinks:                0

Replicated Blocks:
 Total size:    2157927 B
 Total files:   4
 Total blocks (validated):      4 (avg. block size 539481 B)
 Minimally replicated blocks:   4 (100.0 %)
 Over-replicated blocks:        0 (0.0 %)
 Under-replicated blocks:       4 (100.0 %)
 Mis-replicated blocks:         0 (0.0 %)
 Default replication factor:    3
 Average block replication:     1.0
 Missing blocks:                0
 Corrupt blocks:                0
 Missing replicas:              8 (66.666664 %)

Erasure Coded Block Groups:
 Total size:    0 B
 Total files:   0
 Total block groups (validated):        0
 Minimally erasure-coded block groups:  0
 Over-erasure-coded block groups:       0
 Under-erasure-coded block groups:      0
 Unsatisfactory placement block groups: 0
 Average block group size:      0.0
 Missing block groups:          0
 Corrupt block groups:          0
 Missing internal blocks:       0
FSCK ended at Fri Oct 16 16:07:43 BST 2020 in 5 milliseconds
The filesystem under path '/user/student0002/Gutenberg' is HEALTHY
```

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Note

You do not have the required administrative authorities to run Hadoop admin commands. For this reason, you cannot run the command in the following step. The information here is for your reference only.

__ 2. Run the following command:

```
hdfs fsck /
```

The output is similar to the following results:

```
Erasure Coded Block Groups:
 Total size:    0 B
 Total files:   0
 Total block groups (validated):        0
 Minimally erasure-coded block groups:  0
 Over-erasure-coded block groups:       0
 Under-erasure-coded block groups:      0
 Unsatisfactory placement block groups: 0
 Average block group size:      0.0
 Missing block groups:          0
 Corrupt block groups:          0
 Missing internal blocks:       0
FSCK ended at Wed Jul 22 00:30:32 UTC 2020 in 220 milliseconds


The filesystem under path '/' is HEALTHY
```

Information

`hdfs fsck` command is used to check the health of the file system to find missing files, over replicated, under replicated, and corrupted blocks

__ 3. Close all windows.

End of exercise

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Exercise review and wrap-up

In this exercise, you accessed the Hadoop cluster with PuTTY and you explored the Hadoop Hortonworks file structure.

You used basic Hadoop Distributed File System (HDFS) commands to create a new directory and moved some data into HDFS. This data is used in later exercises of this course.

You also used the administrative command `hdfs fsck` to check the health of the file system.

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